

## RANDOM SPLITTINGS: A MODEL FOR A MASS-SIZE DISTRIBUTION

D.M. Schultz and C.F. Crouse

*University of South Africa*

**Key words:** Mass-size distribution, Breakage Process, Logarithmic Gamma distribution, Transition Distribution

### 1. Introduction and Summary

In certain breakage processes when a particle is split into two, it may be assumed that the fraction of the initial mass received by each descendant particle is distributed uniformly on  $(0,1)$ . After  $n$  random splittings of a particle of unit size the mass of the  $n$ -th generation particle is given by a product of the form

$$Z_n = X_1 X_2 \dots X_n$$

where the  $X_i$  are assumed to be independently and uniformly distributed on  $(0,1)$ . The density of  $Z_n$  is given by Feller (1970, p. 26) as

$$(1) \quad f_n(t) = \{1/(n-1)!\} \{n(1/t)\}^{n-1}$$

(Note that in equation (8.2), p. 26 of Feller (1970) the factor  $1/(n-1)$  should read  $1/(n-1)!$ ).

For  $n > 1$  the density (1) is a reverse J-shaped distribution tending to infinity as  $t$  tends to zero.

The well-known asymptotic result of lognormality of a mass-size distribution after a large number of steps in the breakage process was derived through the same type of reasoning as that leading to (1). Authors like Kolmogorov (1941), Epstein (1947) and Halmos (1944) all reached the same asymptotic result but failed to give any so-called transition distribution, i.e., a distribution of mass-size before the asymptotic model provides a good fit.

In practice the distribution of particles in a breakage process plays a very important role in certain standard drum tests of mineral ore, blast furnace coke, etc. After such a test the mill product is sieved into various size intervals, thus producing a mass-size histogram as in Figure 1 below. This distribution clearly cannot be represented by a reverse J-shaped distribution and the reason is obvious. Equation (1) gives the distribution of the mass of an  $n$ -th generation particle, and not the distribution of the masses of all the  $2^n$  particles formed when an initial particle of unit size is split up through  $n$  steps, and

this last distribution is the one most often searched for in practice. The need for a model adequately describing such a mass size distribution is demonstrated, e.g. by Wallach and Sichel (1963) who obtained a distribution which gives a good fit to the mass-size distribution of blast furnace coke after a drum test. Their model, however, does not follow from any logical breakage process as did equation (1) above.

The distribution of mass-size when all particles in the breakage process are taken into account should obviously be different from that in (1) since there is an interdependence between particles at any stage in the process. So for example, after the first splitting of a particle of unit size the masses received by the two descendant particles may be represented by  $x_1$  and  $(1-x_1)$  respectively, where  $x_1$  is an observation on  $X$  which is distributed uniformly on  $(0,1)$ , and these two are clearly not independent. In this paper the distribution of the masses of all the  $2^n$  particles is derived when the breakage process is of the nature described above. This bell-shaped distribution is fitted to a set of observations by the method of moments.

## 2. A Model for describing the distribution of mass

As a first step in deriving a model consider, for example, the third-generation particles formed when a particle of unit mass and initial size  $W$  splits in two as a result of a first collusion, and then each of the two second-generation particles formed again splits in two as a result of second and third collusions. If we denote by  $Y_2$  the random variable of mass-size after these two steps in the breakage process we may write down the following conditional probability mass function of  $Y_2$  given  $x_1, x_2$  and  $x_3$ :

$$(2) \quad P_{Y_2|x_1, x_2, x_3}(y_2) = \begin{cases} x_1 x_2 & \text{if } y_2 = x_1 x_2 W \\ x_1(1-x_2) & \text{if } y_2 = x_1(1-x_2)W \\ (1-x_1)x_3 & \text{if } y_2 = (1-x_1)x_3 W \\ (1-x_1)(1-x_3) & \text{if } y_2 = (1-x_1)(1-x_3)W \\ 0 & \text{elsewhere.} \end{cases}$$

Here  $x_1 x_2$  is the relative mass of a particle of size  $x_1 x_2 W$ , etc., and the  $x_i$  are considered independent observations of the random variable  $X$  which is uniformly distributed on the interval  $(0,1)$ . The moment generating function of  $Y_2$  may now be written as

$$\begin{aligned} M_{Y_2}(\lambda) &= E_{Y_2} \left[ e^{Y_2 \lambda} \right] \\ &= E_X \left[ E_{Y_2|X=x_1, x_2, x_3} (e^{Y_2 \lambda}) \right] \end{aligned}$$

$$= E_X \left[ x_1 x_2 e^{x_1 x_2 W \lambda} + x_1 (1-x_2) e^{x_1 (1-x_2) W \lambda} \right. \\ \left. + (1-x_1) x_3 e^{(1-x_1) x_3 W \lambda} + (1-x_1)(1-x_3) e^{(1-x_1)(1-x_3) W \lambda} \right]$$

and because of symmetry we may write

$$(3) \quad M_{Y_2}(\lambda) = 4E_X \left[ x_1 x_2 e^{x_1 x_2 W \lambda} \right] \\ = \int_0^1 \int_0^1 e^{x_1 x_2 W \lambda} 2x_1 2x_2 dx_1 dx_2$$

Hence it follows that  $Y_2$  is distributed as a product, say  $Z_1 Z_2 W$ , where  $Z_1$  and  $Z_2$  are independently and identically distributed with density function

$$(4) \quad h_Z(z) = 2z, 0 < z < 1.$$

Proceeding in the same way it is clear that  $Y_n$ , the random variable of mass-size of the  $2^n$  particles of the  $n$ -th generation is distributed as a product  $Z_1 Z_2 \dots Z_n W$  where the  $Z_i$  are independently and identically distributed as in (4).

From an application of the central limit theorem to the logarithm of this product it follows that  $\ln Y_n$  is asymptotically distributed as a normal distribution, or that  $Y_n$  is lognormally distributed for large  $n$ . Also, for any value of  $n$ , we may state, after passing to logarithms, that  $-\ln(Y_n/W)$  is distributed as  $\sum_{k=1}^n (-\ln Z_k)$ , where

$$F_{-\ln Z_k}(t) = P[-\ln Z_k \leq t] \\ = 1 - P[Z_k \leq e^{-t}] \\ = 1 - \int_0^{e^{-t}} 2z dz \\ = 1 - e^{-2t}$$

so that

$$(5) \quad f_{-\ln Z_k}(t) = 2e^{-2t}, \quad t > 0$$

Hence  $-\ln(Y_n/W)$  is distributed as the sum of  $n$  independent exponential variates, each with density given by (5). It follows from Feller (1970), p. 25, that

$$g_{-\ell n(Y_n/W)}(t) = 2 \frac{(2t)^{n-1}}{(n-1)!} e^{-2t}$$

$$g_{Y_n/W}(t) = t^{-1} g_{-\ell n(Y_n/W)}(\ell n t^{-1})$$

$$= \frac{2^n}{(n-1)!} t (\ell n \frac{1}{t})^{n-1}$$

$$f_{Y_n}(y_n) = \frac{1}{W} g_{Y_n/W}(y_n/W)$$

$$(6) = \frac{2^n}{(n-1)!} \frac{y_n}{w^2} (\ell n W - \ell n Y_n)^{n-1}$$

This is the distribution of  $Y_n$ , the random variable of mass-size, after a particle of initial size  $W$  went through  $n$  steps in a breakage process.

The distributions (1) and (6) are both logarithmic gamma distributions. The difference between them is more clearly illustrated after transforming to a logarithmic scale. Thus we have from (1)

$$h_{-\ell n} T(u) = \frac{1}{\Gamma(n)} u^{n-1} e^{-u}, \quad 0 < u < \infty,$$

and from (6)

$$g_{-\ell n(y_n/W)}(x) = \frac{2}{\Gamma(n)} (2x)^{n-1} e^{-2x}, \quad 0 < x < \infty.$$

The difference in the second parameter, 2, of the latter distribution is obvious from a comparison of (5) with equation (8.1) on p. 25 of Feller (1970).

The distribution (6) may be regarded as a transition distribution before the lognormal distribution gives a good fit to the mass-size distribution of particles. Convergence to the lognormal form is quite rapid and for  $n$  as large as four or five this limiting form may already be used.

The model in (6) may be made more flexible by characterizing the mode of splitting in a more general way. Instead of assuming  $X$  to be uniformly distributed on the interval  $(0,1)$ , any other symmetric distribution on this interval may be postulated. Consider, e.g., a beta-distribution with equal parameters  $m$ ;

$$f_X(x) = \frac{\Gamma(2m)}{\Gamma^2(m)} x^{m-1} (1-x)^{m-1}, \quad 0 < x < 1, \quad m > 0$$

The value of  $m$  may now be considered a characteristic of the material in the breakage

process since this will determine whether the material more often splits in two more or less equal parts, or whether more often only a fraction of a particle is split off. For this characterization of the breakage process it is shown by Schultz (1971) that after  $n$  steps in the breakage process the random variable  $-\ln(Y_n/W)$  is distributed as the sum of  $m$  gamma variates, each with parameters  $n$  and  $\lambda_i$ , where  $\lambda_1 = m+1$ ,  $\lambda_2 = m+2, \dots, \lambda_m = 2m$ .

### 3. Moments of the distribution

When fitting the distribution (6) in practice the data is usually in the form of a mass-size histogram as in figure 1 below where the total mass of all particles retained by each of a series of successive sieves is recorded. When a total mass  $m_i^*$  is recorded for class interval  $i$ ,  $i=1, 2, \dots, k$ , for which the midpoint is  $u_i$ , we define the  $r$ -th sample moment about the origin as

$$(7) \quad t_r^* = \sum_{i=1}^k m_i^* u_i^r.$$

For a breakage process on a particle of initial size  $W$  after  $n$  steps,  $t_r^*$  is the sample equivalent of the  $r$ -th population moment, so that

$$\begin{aligned} E(t_r^*) &= E[(Z_1 Z_2 \dots Z_n W)^r] \\ &= W^r \prod_{i=1}^n \int_0^1 Z_i \cdot 2Z_i dZ_i \\ (8) \quad &= \left(\frac{2}{r+2}\right)^n W^r \end{aligned}$$

It further follows from Schultz (1971), p. 74, that the variance and covariance of the sample moments  $t_r^*$  are

$$\begin{aligned} (9) \quad v_2(t_r^*) &= W^{2r} \left[ (2\mu'_{2r+2})^n - (2\mu'_{r+1})^{2n} \right. \\ &\quad \left. - 2\mu'_{r+1} \mu'_{r+1} \frac{(2\mu'_{r+1})^{2n} - (2\mu'_{2r+2})^n}{(2\mu'_{r+1})^2 - (2\mu'_{2r+2})} \right] \\ (10) \quad v_{1,1}(t_r^*, t_s^*) &= W^{r+s} \left[ (2\mu'_{r+s+2})^n - (2\mu'_{r+1})^n (2\mu'_{s+1})^n \right. \\ &\quad \left. + 2\mu'_{r+s+2} \frac{(2\mu'_{r+s+2})^n - (2\mu'_{r+1})^n (2\mu'_{s+1})^n}{2\mu'_{r+s+2} - 2\mu'_{r+1} \cdot 2\mu'_{s+1}} \right] \end{aligned}$$

where  $\mu'_r = E[X^r]$

$$\text{and } \mu'_{r,s} = E[X^r(1-X)^s]$$

and where  $X$  is, for the purpose of this paper, assumed to be uniformly distributed on the interval  $(0,1)$ . The limiting distributions of the moments  $t_r^*$  are in general not normal (see Schultz (1971), p. 63).

In practice it seldom happens that the mass-size distribution under consideration has been generated by a breakage process on a single particle. In some standard drum tests carried out on crushed stone, for example, a large number of particles of equal size are initially loaded into the drum. Let us therefore assume that  $p$  particles of size  $W$  were initially in the breakage process. Now the moments  $t_r^*$  cannot be obtained explicitly but we can calculate, from the observed mass-size distribution, the moments

$$t_r = \frac{1}{p} \sum_{j=1}^p t_{r,j}^*$$

$$= \frac{1}{p} \sum_{j=1}^p \sum_{i=1}^k m_{ij}^* u_i^r$$

$$(11) = \frac{1}{p} \sum_{i=1}^k m_i u_i^r$$

where  $m_i = \sum_{j=1}^p m_{ij}^*$  is the total mass in the  $i$ -th class interval due to the joint breakage product of all the  $p$  initial particles. Now it is clear that, for fixed  $n$  and large  $p$ , the well-known asymptotic results will hold for the moments  $t_r$  and functions of these. In particular

$$E[t_r] = E[t_r^*]$$

$$(12) \quad \text{var}(t_r) = \frac{1}{p} \text{var}(t_r^*)$$

$$\text{covar}(t_r, t_s) = \frac{1}{p} \text{covar}(t_r^*, t_s^*)$$

and the moment  $t_r$  will have an asymptotic normal distribution. For the special case discussed in this paper where  $X$  has a uniform distribution it follows from (9), (10) and (12) that

$$\text{var}(t_1) = (W^2/2p) \left[ \left( \frac{2}{3} \right)^{2n} - \left( \frac{2}{5} \right)^n \right]$$

$$(13) \quad \text{var}(t_2) = (7W^4/5p) \left[ \left( \frac{2}{7} \right)^n - \left( \frac{1}{2} \right)^{2n} \right]$$

and  $\text{covar}(t_1, t_2) = 0$ .

#### 4. Estimating the parameters

Estimates of the parameters of the mass-size distribution (6) may be obtained by equating the observed sample moments (11) to their respective expected values (8). So, for example, we may write

$$(14) \quad t_1 = \left(\frac{2}{3}\right)^{\hat{n}} \hat{W}$$

$$t_2 = \left(\frac{1}{2}\right)^{\hat{n}} \hat{W}^2$$

where  $\hat{n}$  and  $\hat{W}$  are the moment estimators of the parameters  $n$  and  $W$ . If we denote the solutions for the two parameters by

$$(15) \quad \hat{n} = \hat{n}(t_1, t_2)$$

$$\hat{W} = \hat{W}(t_1, t_2)$$

it follows from Cramér (1951), p. 366, that these solutions are, for large  $p$ , approximately normally distributed with

$$(16) \quad E(\hat{n}) = n \text{ and } E(\hat{W}) = W$$

and

$$(17) \quad \text{var}(\hat{n}) = \text{var}(t_1) \left( \frac{\partial \hat{n}}{\partial t_1} \right)^2 + \text{var}(t_2) \left( \frac{\partial \hat{n}}{\partial t_2} \right)^2 + \text{covar}(t_1, t_2) \left( \frac{\partial \hat{n}}{\partial t_1} \right) \left( \frac{\partial \hat{n}}{\partial t_2} \right)$$

$$\text{var}(\hat{W}) = \text{var}(t_1) \left( \frac{\partial \hat{W}}{\partial t_1} \right)^2 + \text{var}(t_2) \left( \frac{\partial \hat{W}}{\partial t_2} \right)^2 + \text{covar}(t_1, t_2) \left( \frac{\partial \hat{W}}{\partial t_1} \right) \left( \frac{\partial \hat{W}}{\partial t_2} \right)$$

The solutions in (15) cannot be written explicitly in  $\hat{n}$  and  $\hat{W}$  and consequently the differentials needed in (17) cannot be directly obtained. However, the two equations (14) may be denoted by

$$f_i(\hat{n}, \hat{W}, t_1, t_2) = 0, \quad i=1, 2$$

so that the following differentials may be written down:

$$\frac{\partial f_i}{\partial t_j} + \frac{\partial f_i}{\partial \hat{n}} \frac{\partial \hat{n}}{\partial t_j} + \frac{\partial f_i}{\partial \hat{W}} \frac{\partial \hat{W}}{\partial t_j} = 0, \quad i=1, 2; j=1, 2.$$

By differentiating with respect to  $t_1$ , e.g., we find

$$-1 + \hat{W}(\frac{2}{3})^{\hat{n}} \ln(\frac{2}{3}) \frac{\partial \hat{n}}{\partial t_1} + (\frac{2}{3})^{\hat{n}} \frac{\partial \hat{W}}{\partial t_1} = 0.$$

$$0 + \hat{W}^2(\frac{1}{2})^{\hat{n}} \ln(\frac{1}{2}) \frac{\partial \hat{n}}{\partial t_1} + 2(\frac{1}{2})^{\hat{n}} \hat{W} \frac{\partial \hat{W}}{\partial t_1} = 0$$

from which it follows that

$$\begin{aligned} \frac{\partial \hat{n}}{\partial t_1} &= 2/[(\frac{2}{3})^{\hat{n}} \hat{W} \ln(\frac{8}{9})] \\ (18) \quad \frac{\partial \hat{W}}{\partial t_1} &= (\ln 2)/[(\frac{2}{3})^{\hat{n}} \ln(\frac{8}{9})] \end{aligned}$$

In the same way, after differentiating with respect to  $t_2$ , we find

$$\begin{aligned} \frac{\partial \hat{n}}{\partial t_2} &= -2^{\hat{n}}/[\hat{W}^2 \ln(\frac{8}{9})] \\ (19) \quad \frac{\partial \hat{W}}{\partial t_2} &= [2^{\hat{n}} \ln(\frac{2}{3})]/[\hat{W} \ln(\frac{8}{9})] \end{aligned}$$

Replacing (13), (18) and (19) into (17) gives the required variances.

## 5. A Numerical Example

In a study on the efficiency of an experimental ball mill, the mill was loaded with 1.5 kg of crushed stone of sizes between 1.0 and 1.4 mm. After operating the mill for 20 seconds a sample of approximately 158 g was taken from the contents of the mill through the use of a sample divider. A sieve analysis of this sample was carried out and the mass retained by each sieve recorded. All the crushed stone was now put back into the mill and the whole process repeated for mill periods of 40 seconds, 1, 2, 4 and 8 minutes. By counting a sample of the crushed stone before milling it was determined that 158 g represented approximately 55 300 particles. From the mass-size distributions obtained the values of  $\hat{n}$ ,  $\hat{W}$ ,  $\partial \hat{n}$  and  $\partial \hat{W}$  were calculated from equations (14) and (17) above, using also equations (13), (18) and (19). The results are shown in Table 1 below. The mass-size distribution for the breakage product after 8 minutes of milling is shown in figure 1 together with the density function (5) with the values of  $n$  and  $W$  as in the last row of table 1.

It should be mentioned here that we studied the distribution of particles resulting from *breakage* only. In any process such as that described above an *abrasion* product is usually also formed consisting of very fine particles, mostly in the subsieve range. This

product is usually assumed to be lognormally distributed (Wallach and Sichel (1963)) and produces a second mode near zero in the observed mass-size distribution. In the experiment described here a correction was made in the first class interval of the distribution shown in Figure 1 so that only the breakage product remained for study.

Table 1  
Parameter estimates for the mass-size distribution  
of crushed stone

| Mill time<br>(minutes) | $\hat{n}$ | $\hat{\sigma}_{\hat{n}}$ | $\hat{W}$<br>(mm) | $\hat{\sigma}_{\hat{W}}$<br>(mm) |
|------------------------|-----------|--------------------------|-------------------|----------------------------------|
| $\frac{1}{3}$          | 0,311     | 0,013                    | 1,258             | 0,006                            |
| $\frac{2}{3}$          | 0,526     | 0,017                    | 1,300             | 0,008                            |
| 1                      | 0,719     | 0,019                    | 1,338             | 0,010                            |
| 2                      | 1,157     | 0,024                    | 1,413             | 0,013                            |
| 4                      | 1,750     | 0,030                    | 1,481             | 0,017                            |
| 8                      | 2,477     | 0,036                    | 1,457             | 0,020                            |

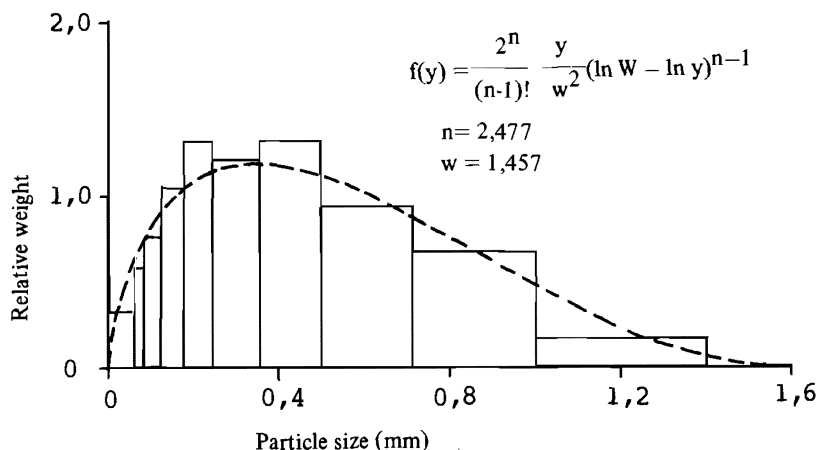


Figure 1 – Mass-size distribution of crushed stone after 8 minutes mill time

The value of the parameter  $n$ , as we would expect, increased very significantly as the breakage process continued for longer periods of time. The last value,  $\hat{n} = 2,477$ , may be interpreted to indicate that all the initial particles went through at least two steps in

the breakage process while about half the third-generation particles already split up when the mill stopped.

We would expect the value of the parameter  $W$  to be a constant just below 1,4 mm. The fact that the experimental results show a variation in the estimate of  $W$  that is just significant may be due to a sensitivity of the method of moments and is at present under investigation.

## ACKNOWLEDGEMENT

The authors are indebted to the South African Council for Scientific and Industrial Research, and to Mr W.J. Taute of the Department of Electrical Engineering of the University of Pretoria for permission to use the experimental results reported above.

## REFERENCES

- CRAMÉR, H. (1951), *Mathematical Methods of Statistics*. Princeton Univ. Press.
- EPSTEIN, B. (1947), The Mathematical Description of certain breakage mechanisms leading to the logarithmico-normal distribution. *Franklin Institute Journal*, **224**, p. 471.
- FELLER, W. (1970), *An Introduction to Probability Theory and its Applications*, Vol. II, 2nd Edition, John Wiley & Sons, Inc.
- HOLMES, P.R. (1944), Random Alms. *Ann. Math. Statist.*, **XV**, p. 182.
- KOLMOGOROFF, A.N. (1941). Über das logarithmisch normale verteilungsgesetz der dimensionen der Teilchen bei Zerstüchelung. *C.R. Acad. Sci. U.R.S.S.*, **31**, p. 99.
- SCHULTZ, D.M. (1971), 'n Wiskundig-Statistiese Model vir 'n Brekingsproses. Unpublished Ph.D. Thesis, University of South Africa.
- WALLACH, R.V. and SICHEL, H.S. (1963). The measurement of mechanical strength and abrasability of Coke by means of Statistical Parameters., *Jnl of Inst. of Fuel*, **36**, 421-435.